

Step-Indexed Hoare Logic as Ramified Forcing

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Abstract

We show that step-indexed Hoare logic is, conceptually and technically, *ramified forcing* — Cohen’s original forcing construction stratified by a well-founded hierarchy of levels. The “later” modality $\triangleright$ is ramification’s level-descent operator; Löb’s rule, the central technical device of step-indexed recursive reasoning, is precisely well-founded induction on the forcing order, internalized into the logic. We make the correspondence precise via a Rocq mechanization that develops the forcing-style propositional semantics, builds Hoare logic over IMP and over an untyped λ -calculus with general recursion, gives an abstract framework parametric in the forcing notion, proves that the well-foundedness is essential by exhibiting a non-well-founded counterexample over $\mathbb{Z}$ where Löb’s rule fails, and formalizes the bridge to the Jaber–Tabareau–Sozeau forcing translation of CIC via a Heyting-algebra-homomorphism embedding $\text{Prop} \hookrightarrow \text{iProp}$. A central methodological observation is that for higher-order recursion the natural soundness proof goes through the internal Löb rule directly, whereas for first-order IMP the same content can be obtained either via the internal rule or via meta-level induction on the step index. We mechanize both forms side by side to make the equivalence concrete. The complete development is approximately 800 lines of Rocq across 11 files, with no external dependencies.

CCS Concepts: • Theory of computation $\rightarrow$ Modal and temporal logics; Program semantics; Hoare logic; Type theory.

Keywords: step-indexed semantics, forcing, ramified forcing, Hoare logic, Löb’s rule, Rocq mechanization, Gödel–Löb logic, forcing translations

1 Introduction

Step-indexed reasoning has become the standard semantic technique for verifying higher-order programming languages with mutable state, recursive types, and concurrency. From Appel and McAllester’s original step-indexed model of recursive types [Appel and McAllester 2001] to the modern Iris ecosystem [Jung et al. 2018], the same basic idea recurs: instead of asking whether a program satisfies a specification, ask whether it satisfies the specification *at every finite observation depth* n , then recover the unindexed statement as the limit. Each level of the hierarchy of approximations is defined without circularity by appeal only to lower levels, and a “later” modality $\triangleright$ mediates the descent from one level to the next.

What the step-indexed literature does not usually say is that this technique is, in classical set-theoretic terms, *forcing*.

Cohen introduced forcing in 1963 to prove the independence of the continuum hypothesis [Cohen 1963, 1966]. In its original presentation — called *ramified forcing* after the ramified type theory of Russell and Whitehead [Russell and Whitehead 1913] — the construction of the generic extension is stratified by a well-founded hierarchy of *levels*, with recursion at level α permitted to consult only levels strictly below α . Shoenfield’s 1971 reformulation [Shoenfield 1971] eliminated the ramification for the set-theoretic application: the modern textbook presentation of forcing does not use it. But the structure — well-founded stratification, level descent, a forcing relation $p \Vdash \varphi$ that is monotone in the strength of p — is exactly the structure of step-indexed logic.

In this paper we make the correspondence precise. The forcing *conditions* of step-indexed Hoare logic are the natural numbers ω with the order $\leq$; the forcing *relation* $n \Vdash \varphi$ is monotone in n (downward closure); the *later* modality $\triangleright$ is the level descent operator of ramified forcing; and Löb’s rule, which underwrites all recursive reasoning in step-indexed systems, is precisely well-founded induction on the forcing order, internalized into the logic.

Why does this matter? The forcing reading is not just a re-labelling. It clarifies three things that are otherwise scattered across the step-indexed literature.

First, it explains *why* the technique has the shape it does. The well-founded stratification is not a presentation convenience to be eliminated; it is constitutive of the soundness of the modal logic. Section 7 makes this precise: we exhibit a non-well-founded forcing notion (the integers $\mathbb{Z}$) where every other piece of the framework goes through, but Löb’s rule fails.

Second, it diagnoses where the internal Löb rule earns its keep. Section 5 compares two mechanized soundness proofs of recursive Hoare rules: the while rule for IMP (Section 3) and the fix rule for the untyped λ -calculus (Section 4). The first admits an unproblematic meta-level proof by induction on the step index; the second does not, and the natural proof invokes the internal Löb rule directly. The structural reason — that the soundness predicate is itself recursive in the fix case — is made precise by the forcing reading.

Third, it identifies step-indexed Hoare logic as the propositional fragment of an existing construction in the type-theory literature: the Jaber–Tabareau–Sozeau forcing translation of CIC [Jaber et al. 2012]. Section 8 formalizes this identification via a constant embedding $\text{Prop} \hookrightarrow \text{iProp}$ that is a Heyting algebra homomorphism, with $\triangleright$ appearing as the step-indexed-specific addition.

Contributions.

1. A mechanized re-presentation of step-indexed Hoare logic as ramified forcing, in Rocq. The propositional layer (Section 2) gives a forcing semantics for an intuitionistic logic with the later modality and Löb's rule. The Hoare layer is constructed over IMP (Section 3) and over an untyped λ -calculus with general recursion (Section 4).
2. An explicit comparison of the IMP while rule and the λ -calculus fix rule that diagnoses where the internal Löb rule is essential (Section 5). For IMP, soundness goes through either via meta-level induction on the step index or via the internal Löb rule; we exhibit both proofs and observe their structural identity. For the λ -calculus with fix, the natural soundness proof must invoke the internal Löb rule, because the soundness predicate is itself recursive in the program. This is the strongest piece of evidence for the slogan that "Löb is what makes step-indexed Hoare logic work in higher-order settings."
3. An abstract forcing framework (Section 6) parametric over an arbitrary well-founded forcing structure. We exhibit two instances: $(\omega, \leq)$, recovering the standard step-indexed semantics; and $(\omega \times \omega, \leq_{pw})$, with the pointwise order, modelling independent step-indices for parallel components. The internal Löb rule, in the abstract setting, is proved by `well_founded_ind` on the strict refinement order. This makes precise the equivalence: *internal Löb is well-founded induction on the forcing order, internalized*.
4. A small new metatheoretic result: *well-foundedness of the forcing notion is essential* (Section 7). We exhibit a non-well-founded "would-be" forcing structure over $\mathbb{Z}$, in which every `iProp` construction goes through except Löb's rule. The premise of Löb's rule $(\triangleright \perp) \rightarrow \perp$ is forced at every condition — vacuously, since $\mathbb{Z}$ has no minimum — while $\perp$ is forced at none. This is a special case of the model theory of the Gödel–Löb provability logic [Boolos 1993; Solovay 1976], brought into the step-indexed Hoare logic setting and mechanized.
5. A partial bridge to the Jaber–Tabareau–Sozeau forcing translation of CIC [Jaber et al. 2012] (Section 8). We formalize the constant embedding `Prop` $\hookrightarrow$ `iProp` as a Heyting algebra homomorphism, and observe that the $\triangleright$ modality is the genuinely new content of the `iProp` universe over the embedded image of `Prop`. A full mechanization of the type-level JTS translation is future work.

What is and is not new. The individual ingredients are largely known to specialists in neighbouring fields. The topos-of-trees perspective on step-indexing [Birkedal et al. 2012] is implicitly a forcing semantics; the Gödel–Löb modal logic of provability is the modal logic of provability and well-foundedness; the JTS forcing translation of CIC is the

type-theoretic analog of forcing. What is new is the assembly: a single mechanized account that makes the forcing reading explicit, diagnoses the role of the internal Löb rule, and gives a parametric framework with the metatheoretic guarantee that well-foundedness cannot be relaxed.

Outline. Section 2 gives the propositional layer of the forcing semantics, including the Heyting-algebra connectives, the later modality, and Löb's rule. Section 3 develops Hoare logic for IMP and proves the while rule by meta-level induction; Section 4 develops Hoare logic for an untyped λ -calculus with fix and proves the soundness rule for fix via the internal Löb rule. Section 5 compares these two proofs explicitly and articulates the diagnostic claim. Section 6 gives the abstract forcing framework. Section 7 proves Löb's rule fails without well-foundedness. Section 8 establishes the constant-embedding bridge to JTS forcing translations. Section 9 positions the contribution against the literature. Section 10 concludes.

Rocq development. The complete Rocq development is approximately 800 lines across 11 files, building cleanly against the Rocq standard library only (no external libraries such as Iris or stdpp).

2 The Propositional Layer

[TBD: forcing semantics for propositional intuitionistic logic with the later modality and Löb's rule. Covers: conditions as $\mathbb{N}$ with $\leq$; `iProp` as downward-closed predicates with monotonicity; Heyting connectives $\top, \perp, \wedge, \vee, \rightarrow$; the later modality $\triangleright$; Löb's rule $(\triangleright \phi \rightarrow \phi) \vdash \phi$, proved by induction on the step index; the sanity check that $\triangleright \perp \not\vdash \perp$. References Phase 1 of the Rocq development.]

3 Hoare Logic for IMP

[TBD: the first-order calibration setting. Covers the IMP language and its small-step semantics; the step-indexed `wp_at` predicate and the Hoare triple as an `iProp`; soundness of skip, assignment, sequencing, conditional, and consequence. Then the while rule, proved by induction on the step index, and an alternative proof invoking the internal Löb rule via `ientails_trans` and `iLob`. References Phases 2a/2b/2c.]

4 Hoare Logic for an Untyped λ -Calculus with Recursion

[TBD: the higher-order setting where step-indexing becomes essential. Covers the untyped λ -calculus with fix (de Bruijn syntax, CBV small-step semantics with β and fix-unfolding); the step-indexed `wp_at` with an explicit safety conjunct (because the λ -calculus has stuck terms); the `wp_value` and `wp_pure_step` rules. The `wp_fix` soundness rule, proved by apply `ientails_trans` to reduce to the form $\top \vdash \triangleright \phi \rightarrow \phi$, then apply `iLob`. No meta-level induction. The worked example `recursive_id` (a

fix-based identity) and bottom (canonical divergent fix-point).
References Phases 3a/3b/3c/3d.]

5 Why the Two Proofs Look Different

[TBD: the central diagnostic section. Compares the two soundness proofs of Sections 3 and 4 side by side. In IMP, the wp predicate for while reduces to an induction on the step count without further recursion in the predicate itself. In the λ -calculus with fix, the wp predicate for fix is intrinsically recursive in the self-reference, and unrolling the recursion at the meta level requires the (well-founded) induction principle that Löb's rule internalizes. The structural diagnostic: when the soundness predicate is itself recursive in the program, internal Löb is the natural tool; otherwise it is interchangeable with outer induction on the step index.]

6 An Abstract Forcing Framework

[TBD: Phase 4. The ForcingStructure record packaging a preorder $\leq$, a strict relation $<$, the standard order axioms, and well-foundedness of $<$. Over an arbitrary ForcingStructure, the iProp, Heyting connectives, and the later modality $\triangleright\phi$ defined as universal-over-strict-predecessors: $p \Vdash \triangleright\phi$ iff $\forall p' < p, p' \Vdash \phi$. Löb's rule proved by well_founded_ind on $<$. Two instances: nat_FS (the standard $(\omega, \leq)$ recovering Phase 1) and prod_FS (the pointwise product $\omega \times \omega$). The conceptual claim: at the meta level, internal Löb's rule is well-founded induction on the forcing order, viewed internally.]

7 Well-foundedness is Essential

[TBD: Phase 5. The metatheoretic observation that the fs_lt_wf axiom in ForcingStructure cannot be dropped. We construct a “would-be” iProp framework over $\mathbb{Z}$, in which all of iPropZ, iImpLZ, iLaterZ go through with their standard definitions. We show: (i) the premise of Löb's rule, $(\triangleright\perp) \rightarrow \perp$, is forced at every condition, because $\mathbb{Z}$ has no minimum and $\text{so}\triangleright\perp$ is forced at no condition; (ii) $\perp$ itself is not forced at every condition; (iii) hence the entailment $(\triangleright\perp \rightarrow \perp) \vdash \perp$ does not hold. This is the step-indexed analog of the modal logic fact that GL is sound on the class of converse-well-founded Kripke frames and not on the integers. The methodological upshot: the “ramification” in ramified forcing is not a presentation convenience; it is constitutive of the modal logic.]

8 Bridge to Forcing Translations

[TBD: Phase 6. The constant embedding $\text{Prop} \hookrightarrow \text{iProp}$ defined as $[P] := \text{MkProp}(\lambda_P) \dots$. We prove it is a Heyting algebra homomorphism: it commutes with $\top$, $\perp$, $\wedge$, $\vee$, $\rightarrow$ in both directions of entailment. The genuine new content of iProp over the embedded image is the $\triangleright$ modality, witnessed by $\triangleright[\text{False}] \not\vdash [\text{False}]$. The framework of Section 6 is therefore the propositional fragment of the Jaber–Tabareau–Sozeau forcing translation of CIC [Jaber et al. 2012], instantiated at

an arbitrary forcing structure. A full mechanization of the type-level JTS translation is future work.]

9 Related Work

This section is organized into seven literature threads: (i) step-indexing in programming language semantics; (ii) forcing in set theory, ramified and unramified; (iii) topos-theoretic models of step-indexing; (iv) forcing translations in type theory; (v) the modal logic of provability (GL) and well-foundedness; (vi) constructive modal logic for guarded recursion; (vii) predicativity and ramified hierarchies in foundations of mathematics. A final subsection states the gap this paper fills.

9.1 Step-indexing in programming language semantics

Step-indexed logical relations were introduced by Appel and McAllester [Appel and McAllester 2001] to give a semantic account of recursive types in the setting of foundational proof-carrying code. The technique gives a meaning to recursive types not via a fixed point construction in some semantic domain, but by approximating the intended meaning at every finite “observation depth” n . Two terms are equivalent at index n if they cannot be distinguished by any context running for at most n steps; the desired equivalence is then the limit (intersection) of all n -equivalences. Appel and McAllester show that this gives a sound model for recursive types and that the resulting logical relation is preserved under reduction — the canonical *step-down* property that we will recognise as the monotonicity of step-indexed propositions.

Ahmed's PhD thesis [Ahmed 2004] extended step-indexing to languages with mutable references, including general references that can store functions of any type (“higher-order store”). This is the classical setting in which step-indexing is essential rather than merely convenient: the standard set-theoretic recursion on type structure is blocked by the circularity that comes from storing higher-order functions in mutable cells. Ahmed shows how the step-indexed approach absorbs this circularity into the index hierarchy.

Birkedal and collaborators have extended these ideas in two directions. In one direction, *topos-theoretic* reformulations (see Section 9.3 below) replaced the explicit step indices with categorical structure. In the other, the technique was scaled up to richer programming languages and verification settings; see Birkedal et al. [Birkedal et al. 2011] on step-indexed Kripke models for capability calculi.

The current state of the art on the programming-languages side is the *Iris* family of separation logics [Jung et al. 2018; Krebbers et al. 2017]. *Iris* extends step-indexed separation logic with higher-order ghost state, invariants, and a powerful set of derived rules. Its soundness theorem is mechanized in Coq against a model built ultimately on step-indexed propositions. The internal logic of *Iris* is itself a step-indexed

modal logic with a “later” modality $\triangleright$ and a Löb rule; substantial libraries of Coq tactics support reasoning within it. Other recent step-indexed systems include VST [Appel et al. 2014] and Transfinite Iris [Spies et al. 2021].

Two technical observations from this literature are essential for our work. First, the “later” modality has been a unifying device since at least Appel et al. [Appel et al. 2007], who introduced it as the very-modal $\triangleright$ for guarded recursion. Second, the soundness proof of every step-indexed logic relies on a Löb-style argument — either as an explicit lemma in the internal logic, or implicitly through induction on the step index in the soundness proof. Our paper takes seriously the question of which formulation is more natural in which setting, and argues that for higher-order recursion the internal Löb form is strictly preferable.

9.2 Forcing in set theory: ramified and unramified

Cohen introduced forcing in his 1963 papers [Cohen 1963, 1964] to prove the independence of the continuum hypothesis from ZFC, and gave a book-length treatment in 1966 [Cohen 1966]. The original presentation is what we will call *ramified forcing*: the construction of the generic extension $M[G]$ is stratified by ordinal levels mimicking Gödel’s constructible hierarchy. A “ramified language” $\mathcal{L}^*$ has a type symbol for each ordinal α below some sufficient bound, and a variable of type α ranges only over sets at levels strictly below α . The forcing relation $p \Vdash_\alpha \varphi$ is defined level by level: forcing of a level- α formula recurses only on forcing at levels strictly below α .

The two features that make this presentation “ramified” are predicative stratification (a level- α formula only quantifies over levels strictly below α) and explicit level descent (the recursive definition of forcing is structured by ordinal predecessor). Both features are inherited from the ramified theory of types of Russell and Whitehead [Russell and Whitehead 1913] and from the broader tradition of predicative analysis [Feferman 1998; Weyl 1918].

Shoenfield’s 1971 lecture [Shoenfield 1971] reformulated forcing without the ramification: forcing conditions and P -names are defined by transfinite recursion on rank, and the forcing relation is defined by recursion on formula complexity, without explicit stratification by syntactic level. Shoenfield’s presentation is mathematically equivalent to Cohen’s — the same generic extensions are produced — but it is simpler, and it has become the standard modern presentation in textbooks [Jech 2003; Kunen 2011]. Boolean-valued models, due to Scott and Solovay [Scott and Solovay 1967], give yet another equivalent presentation.

What is preserved across these reformulations is the underlying *forcing relation* $p \Vdash \varphi$ together with its fundamental properties. What was eliminated — the explicit ramified syntactic hierarchy — is exactly what we will argue is *not* eliminable in the step-indexed setting.

9.3 Topos-theoretic models of step-indexing

The connection between step-indexing and category-theoretic constructions was made explicit by Birkedal, Møgelberg, Schwinghammer, and Støvring [Birkedal et al. 2012] via the *topos of trees*, the presheaf topos on the partially ordered set $(\omega, \leq)$. They observed that step-indexed propositions are the propositions of the internal logic of this topos, and that the “later” modality $\triangleright$ arises naturally as a particular endofunctor on objects of the topos. Their paper introduces *synthetic guarded domain theory*, in which all of the step-indexed semantics is internalized: rather than indexing manually, one works in a type theory whose intrinsic semantics is the topos of trees.

The presheaf-topos perspective makes precise much of what the early step-indexed literature handled informally. In particular, the monotonicity condition on step-indexed propositions corresponds exactly to the presheaf condition (functoriality), and the soundness of the Löb rule is the existence of a fixed point of the $\triangleright X = F(X)$ for a contractive F , or equivalently the well-foundedness of $(\omega, <)$ as the index domain). Subsequent work has refined the categorical setting and used it to model more sophisticated languages [Bizjak et al. 2016].

The presheaf-topos perspective is implicitly a forcing-style semantics in the model-theoretic sense: a presheaf topos on $(P, \leq)$ is exactly the topos of sheaves with respect to the trivial Grothendieck topology on P , and these are exactly the Kripke models of intuitionistic logic with worlds in P . The topos’s truth value object plays the role of a Heyting algebra-valued “forcing” truth value. To our knowledge, this connection has not been emphasized in the topos-of-trees literature, and the word “forcing” has not been prominent in it. Our paper makes this terminological and conceptual connection explicit.

9.4 Forcing translations in type theory

A parallel literature, originating in the constructive type theory community, develops syntactic *forcing translations* of type theories. The seminal work here is Jaber, Tabareau, and Sozeau [Jaber et al. 2012], who give a forcing translation of the Calculus of Constructions: given a forcing notion (a poset P), they exhibit a syntactic translation that takes a CIC term $M : T$ to a forced term $M^P : T^P$, where the type translation $T \mapsto T^P$ corresponds to “the forced version of T .” For propositions, T^P is a downward-closed predicate on P — exactly our iProp of Section 2.

This line was extended in [Jaber et al. 2016] (the “definitional side of the forcing”), which gives a refined translation that preserves more of the computational structure of the original type theory. Coquand and collaborators have related work on constructive presentations of forcing in type-theoretic settings [Coquand and Jaber 2010].

The Jaber et al. translations are the type-theoretic version of ramified forcing for type theory: a parametric construction of a “forced universe” that supports the same operations as the unforced one. Our Phase 4 abstract framework recovers the propositional fragment of this translation by direct construction in Rocq. The Phase 6 file makes the embedding precise via a constant embedding $\text{Prop} \hookrightarrow \text{iProp}$ that is a Heyting algebra homomorphism.

What our work does not do — and what the Jaber et al. line does — is give a translation of the *full* type theory, including dependent types. A natural future direction is to formalize the restriction of the Jaber et al. translation to the propositional fragment in Rocq and check (mechanically) that it produces our iProp up to definitional equality.

9.5 Modal logic of provability and well-foundedness

The Gödel–Löb provability logic GL is the modal logic characterized by the axiom $\Box(\Box\varphi \rightarrow \varphi) \rightarrow \Box\varphi$ (the “Löb axiom”), together with the standard K axioms. Solovay’s arithmetical completeness theorem [Solovay 1976] establishes that GL is sound and complete for arithmetical interpretations of $\Box$ as “it is provable in Peano Arithmetic that...”. From the model-theoretic side, GL is sound and complete for the class of transitive Kripke frames whose accessibility relation R is *converse well-founded* (no infinite ascending chain $w_0 R w_1 R w_2 R \dots$); see Smoryński [Smoryński 1985] and Boolos [Boolos 1993] for textbook treatments.

The model theory of GL is directly relevant to our work. The “later” modality $\triangleright$ of step-indexed logic is formally exactly a GL-style provability $\Box$ once we identify the strict refinement order $\prec$ on conditions with the converse of the GL accessibility relation. Löb’s rule of step-indexed logic is the Löb axiom of GL. The fact that $\triangleright$ requires well-foundedness for Löb’s rule to hold is the dual of GL’s requirement that the accessibility relation be converse well-founded. Our Phase 5 result — that Löb fails over $\mathbb{Z}$ — is therefore a special case of (the converse of) the soundness of GL for converse-well-founded frames.

What our paper adds to this picture is the observation that the familiar PL-style step-indexed logic, including its mechanizations in Iris and elsewhere, is in fact a propositional fragment of GL indexed over a particular kind of GL-frame (one with well-founded strict refinement). This connection is folklore in modal logic but, as far as we are aware, has not been made explicit or mechanized for the step-indexed Hoare logic setting.

9.6 Constructive modal logic and guarded recursion

A separate line of work, with substantial overlap with synthetic guarded domain theory, develops constructive modal logics for guarded recursion. Atkey and McBride [Atkey and McBride 2013] use the $\triangleright$ modality to characterize *productive coprogramming*: a corecursive function is productive iff each recursive call sits under a $\triangleright$. Clouston, Bizjak, Grathwohl,

and Birkedal [Clouston et al. 2015, 2017] develop guarded λ -calculi with $\triangleright$ as a primitive type former, supporting recursive types via a guarded fixed-point operator. Møgelberg and others [Bizjak and Møgelberg 2018; Møgelberg 2014] extend these systems to dependent types and provide categorical semantics.

This literature is closely allied with synthetic guarded domain theory: both treat $\triangleright$ as a built-in modal operator and exploit its compatibility with recursion. The work is structured mostly around *computational* and *type-theoretic* questions (“how do we type recursion productively?”), rather than *semantic* questions (“what does the step-indexed semantics look like as a forcing construction?”). Our paper is complementary: we work on the semantic side, with the forcing reading as our organizing principle.

9.7 Predicativity and ramified hierarchies

The word “ramified” belongs to a tradition older than Cohen’s forcing. Russell and Whitehead [Russell and Whitehead 1913] introduced the ramified theory of types in *Principia Mathematica* to resolve the set-theoretic paradoxes via a stratification of propositions by “order,” where a proposition of order n may quantify only over propositions of order $< n$. The same idea animates Weyl’s predicative analysis [Weyl 1918], in which the real numbers are constructed within an arithmetical hierarchy without appeal to impredicative set-theoretic principles. Feferman’s program of *predicativity given the natural numbers* [Feferman 1998] develops this into a comprehensive foundation for substantial parts of classical mathematics.

The thread connecting ramified type theory to ramified forcing is that both stratify quantification by a well-founded hierarchy, and both treat the hierarchy itself as part of the syntactic apparatus rather than as a meta-level convenience. Shoenfield’s unramification of forcing showed that for forcing the stratification can be absorbed into a single recursion on rank; what we argue, by mechanizing step-indexed Hoare logic in higher-order settings, is that for step-indexing the analogous unramification *cannot* be performed. The well-founded forcing notion is doing real work, not just structuring the proof.

This connection also clarifies why the Iris-style internal logic “feels predicative”: step-indexed propositions are stratified by a well-founded hierarchy, and any recursive definition that does not respect the hierarchy — by mediating recursion through a $\triangleright$ — is rejected. The system is not just predicative in spirit; it is precisely a ramified predicative system in the technical sense.

9.8 Hoare logic and program verification

We omit a survey of classical Hoare logic and separation logic, referring the reader to standard sources [Brookes 2007; Hoare 1969; O’Hearn et al. 2001]. Two points are worth noting. First, our IMP development in Section 3 uses the plain

(non-separation) version of Hoare logic, since separation machinery is not needed for the conceptual point we want to make about the while rule. Second, Cook’s relative completeness theorem [Cook 1978] for Hoare logic shows that the classical proof system is as complete as can be hoped given any reasonable specification language; our results are orthogonal to this — we study how the soundness side is best organized, not the proof system.

9.9 The gap this paper fills

The summaries above suggest that almost every ingredient of our work is, individually, well-known to specialists in some adjacent community. The topos-of-trees perspective on step-indexing is familiar in synthetic guarded domain theory; the GL modal logic of provability is familiar to modal logicians; the ramified-forcing presentation of Cohen is familiar to set theorists; the JTS forcing translation is familiar to the type theory community; and the predicative reading of ramification is familiar to foundations researchers.

What does *not* appear in the literature, to our knowledge, is a single mechanized account of step-indexed Hoare logic that (i) frames step-indexing explicitly as ramified forcing, making the connection to Cohen’s original presentation precise rather than implicit in the topos-theoretic literature; (ii) demonstrates that internal Löb is the load-bearing soundness tool for higher-order recursion by mechanizing both an IMP-style and a λ -calculus-with-`fix` Hoare logic and exhibiting both the meta-level and internal forms of the while rule’s soundness proof; (iii) generalizes the framework to abstract forcing notions parametrically, exhibiting at least one second instance and recovering the standard $(\omega, \leq)$ instance; (iv) proves the well-foundedness of the forcing notion is essential by an explicit counterexample over $\mathbb{Z}$ where the `iProp` constructions go through but Löb’s rule fails; (v) verifies the bridge to the JTS forcing translation by formalizing the constant embedding `Prop` $\hookrightarrow$ `iProp` as a Heyting algebra homomorphism and identifying $\triangleright$ as the step-indexed-specific addition.

The contribution is therefore not the introduction of any single new construction but the *integration* of several existing threads into one mechanized account, plus the small new technical result of Section 7 and the explicit comparison of Sections 3 and 4 that diagnoses where internal Löb earns its keep.

10 Conclusion

We have given a forcing-style semantics for step-indexed Hoare logic, mechanized in Rocq across the propositional layer, an IMP Hoare logic, and an untyped λ -calculus with general recursion. The forcing reading places the step-indexed technique in direct correspondence with Cohen’s ramified forcing construction: the natural numbers are the forcing conditions, the later modality $\triangleright$ is the level-descent operator,

and Löb’s rule is well-founded induction on the forcing order, internalized into the logic.

What we found. Two observations stand out. First, in the first-order setting of IMP, the while rule admits two structurally identical soundness proofs: an outer induction on the step index, and an appeal to the internal Löb rule. The two proofs have the same content because at the meta level, Löb’s rule *is* induction on the step index. Mechanizing both side by side, as we do, makes the equivalence concrete rather than analogical.

Second, in the higher-order setting of the untyped λ -calculus with `fix`, the analogous soundness proof has a different shape. The soundness predicate for a recursive function is itself defined in terms of the function being verified, and unrolling this self-reference at the meta level produces a non-trivial induction that does not arise from the program’s step count alone. The natural proof, which we mechanize, invokes the internal Löb rule directly: “show that under the assumption $\triangleright\varphi$ we have φ , then conclude φ by Löb.” No meta-level induction on a step count is needed. This is, in our opinion, the simplest diagnostic for the slogan that *Löb is what makes step-indexed Hoare logic work for higher-order recursion*.

Three further observations. The abstract framework of Section 6 makes precise that the choice of ω as the step-index domain is a *computational* convenience rather than a foundational one. Any well-founded forcing notion supports the same internal logic, with $\triangleright$ defined uniformly as “forces at every strict predecessor” and Löb derived by well-founded induction. We exhibit the pointwise product $\omega \times \omega$ as a second instance, but the framework is generic.

Section 7 shows that the well-foundedness cannot be dropped: we construct a non-well-founded forcing notion over $\mathbb{Z}$ in which Löb’s rule fails. This is a special case of the model theory of GL, but the statement for step-indexed Hoare logic does not, as far as we are aware, appear in the verification literature. The claim has methodological weight: step-indexed Hoare logic is not just “Hoare logic with a step counter”; it requires a well-founded forcing notion to be sound at all.

Section 8 verifies that the `iProp` framework is the propositional fragment of the Jaber–Tabareau–Sozeau forcing translation of CIC, via a constant embedding `Prop` $\hookrightarrow$ `iProp` that we prove to be a Heyting algebra homomorphism. The $\triangleright$ modality is the genuinely new content of `iProp` over the embedded image, and arises uniformly from the well-founded strict refinement on conditions.

Limitations. The Hoare logic for the λ -calculus in Section 4 is, at present, restricted to partial correctness against a single specification predicate S that serves as both pre- and post-condition. The `fix` soundness rule `wp_fix` (Section 4) accordingly handles only this case. A more general rule with

distinct pre/post-conditions, a rule for fix at higher types, and an integration with separation logic on a heap are all natural directions but lie outside the present development.

The bridge to JTS forcing translations of Section 8 formalizes only the constant embedding. A full mechanization of the type-level translation, against an explicit forcing structure and with the propositional rules verified to match our $iProp$, is the natural next step.

The abstract framework of Section 6 parametrizes the propositional layer over arbitrary well-founded forcing structures, but the IMP and λ -calculus developments of Sections 3–4 are presented at the specific $(\omega, \leq)$ instance for clarity. Lifting them to be parametric in the forcing structure is straightforward and not done here.

Future directions. Several extensions suggest themselves.

- *Mechanizing the full JTS translation* (or a useful fragment) and verifying that its propositional fragment coincides definitionally with our $iProp$ at the corresponding forcing structure. This would convert the heuristic correspondence of Section 8 into a theorem.
- *Step-indexing over richer forcing structures* — transfinite ordinals (cf. Transfinite Iris [Spies et al. 2021]), branching conditions for genuinely concurrent step counters, or mixed structures combining several step-index dimensions. The abstract framework already supports these in principle; the work is to exhibit non-trivial verifications at the new instances.
- *Lifting the Hoare-logic developments to the abstract framework*, so that IMP and the λ -calculus admit verifications parametric in the forcing structure. This is straightforward but bookkeeping-heavy; it would make the parametric story of Section 6 the primary presentation rather than an afterthought.
- *An integration with separation logic* along the lines of Iris, lifting the forcing reading from the propositional logic to the resource-aware extension. This is a substantial undertaking but would let the forcing perspective speak directly to state-of-the-art verification.

The broader point we hope the paper makes is that step-indexed reasoning, viewed as a forcing construction, sits at the intersection of three research traditions that do not usually talk to each other: set-theoretic forcing, modal logic of provability, and step-indexed semantics for verification. Each tradition contains piece-of-the-picture insights that the other two could profit from. The mechanization that accompanies this paper is intended both as evidence for the technical claims and as a starting point for others wishing to explore the intersection further.

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